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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER | | | | | |

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9964901203

MATHEMATICS (SYLLABUS D) **4024/13**

Paper 1 **May/June 2010**

2 hours

Candidates answer on the Question Paper.

Additional Materials: Geometrical instruments

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

If working is needed for any question it must be shown in the space below that question.
Omission of essential working will result in loss of marks.

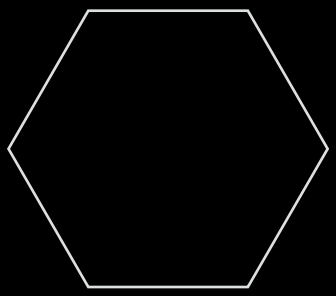
NEITHER ELECTRONIC CALCULATORS NOR MATHEMATICAL TABLES MAY BE USED IN THIS PAPER.

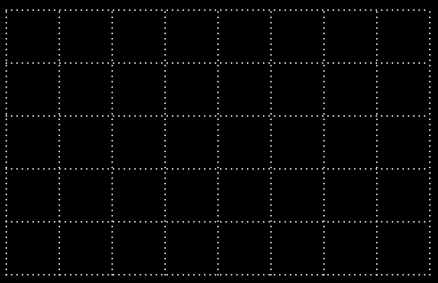
The number of marks is given in brackets [ ] at the end of each question or part question.
The total of the marks for this paper is 80.

For Examiner's Use

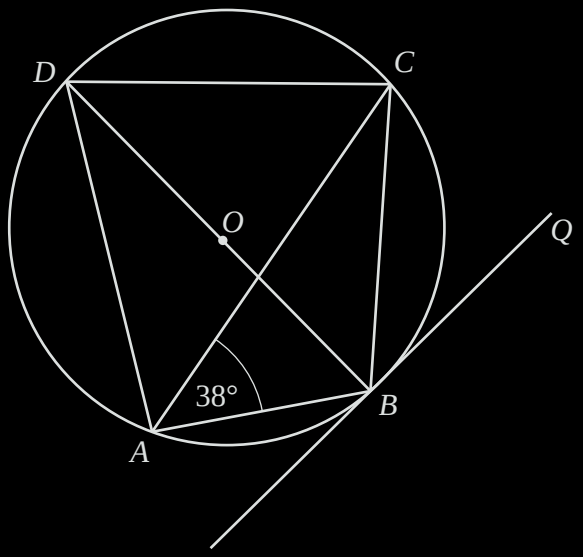
This document consists of **20** printed pages.

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The diagram shows a circle, centre O, passing through A, B, C and D.
BOD is a straight line and $B\hat{A}C = 38°$.
The line BQ is a tangent to the circle at B.

Find

(a) $D\hat{A}C$,

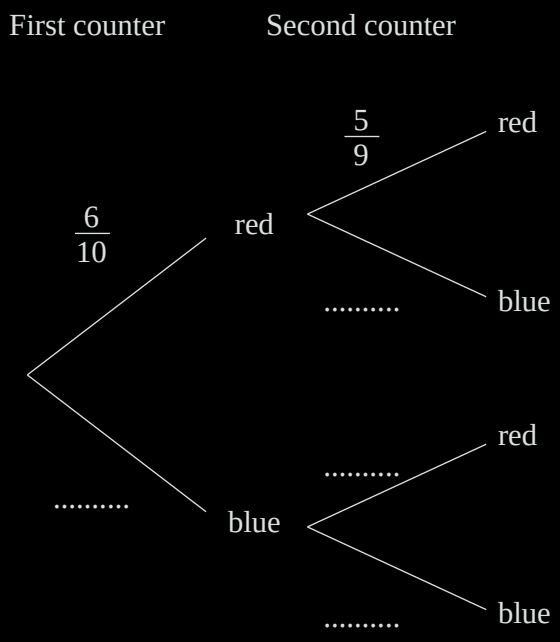
Answer (*a*) $D\hat{A}C$ = ... [1]

(b) $D\hat{B}C$,

Answer (*b*) $D\hat{B}C$ = ... [1]

(c) $C\hat{B}Q$.

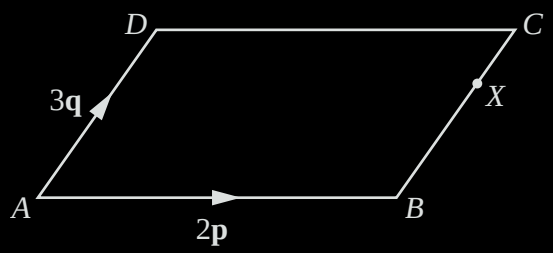
Answer (*c*) $C\hat{B}Q$ = ... [1]



[1]

(b) Expressing your answer as a fraction in its simplest form, calculate the probability that both counters are the same colour.

Answer (*b*) .....[2]



ABCD is a parallelogram.
X is the point on *BC* such that $BX : XC = 2 : 1$.
$\overrightarrow{AB} = 2\mathbf{p}$ and $\overrightarrow{AD} = 3\mathbf{q}$.

Find, in terms of **p** and **q**,

(a) $\overrightarrow{AC}$,

Answer (*a*) $\overrightarrow{AC}$ = ...[1]

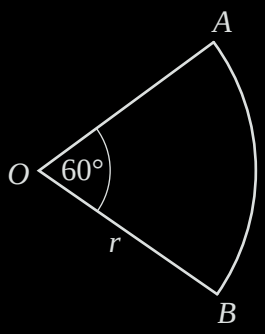
(b) $\overrightarrow{AX}$,

Answer (*b*) $\overrightarrow{AX}$ = ... [1]

(c) $\overrightarrow{XD}$.

Answer (*c*) $\overrightarrow{XD}$ = ... [1]

Find, in its simplest form, an expression in terms of r and π for



(a) the area of the sector,

Answer (*a*) ..... cm^2 [1]

(b) the perimeter of the sector.

Answer (*b*) ..... cm [2]

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19 $\mathbf{A} = \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 0 & 2 \\ -1 & 3 \end{pmatrix}$.

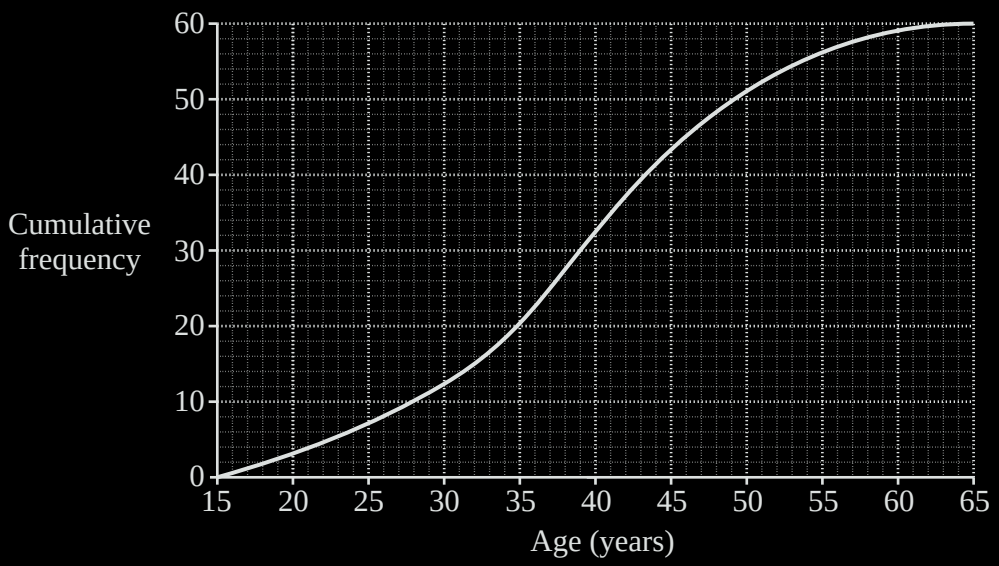
Find

(a) $\mathbf{A} - \mathbf{B}$,

Answer (*a*) $\begin{pmatrix} & \\ & \end{pmatrix}$ [1]

(b) $\mathbf{B}^{-1}$.

Answer (*b*) $\begin{pmatrix} & \\ & \end{pmatrix}$ [2]



Use the graph to estimate

(a) the median,

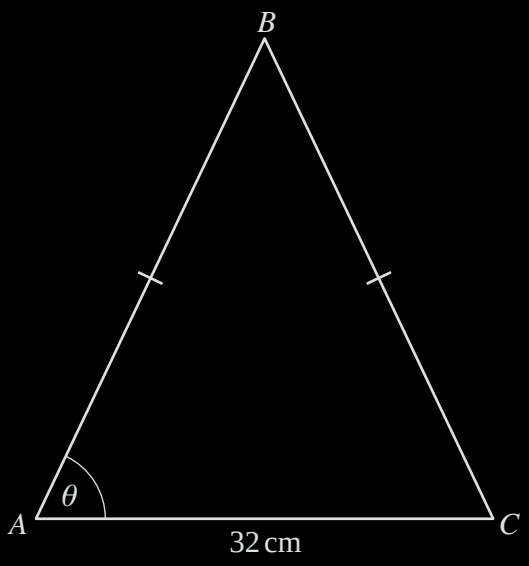
Answer (*a*) ..... [1]

(b) the interquartile range,

Answer (*b*) ..... [2]

(c) the number of employees aged over 50.

Answer (*c*) ..... [1]



sin θ	$\frac{15}{17}$
cos θ	$\frac{8}{17}$
tan θ	$\frac{15}{8}$

ABC is an isosceles triangle with $AB = BC$ and $AC = 32$ cm.

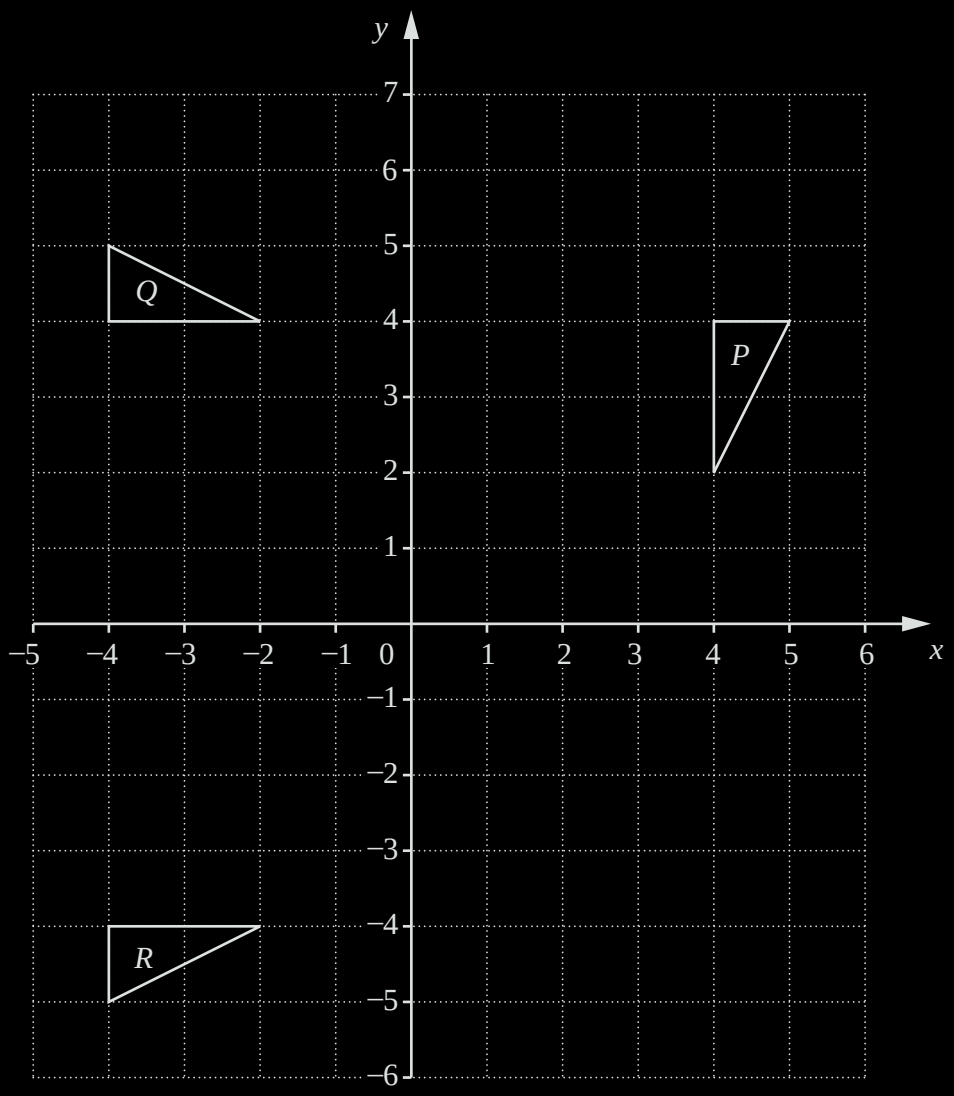
Using as much information from the table as is necessary, calculate

(a) *AB*,

Answer (*a*) ..... cm [2]

(b) the area of triangle *ABC*.

Answer (*b*) ..... cm^2 [2]



[2]

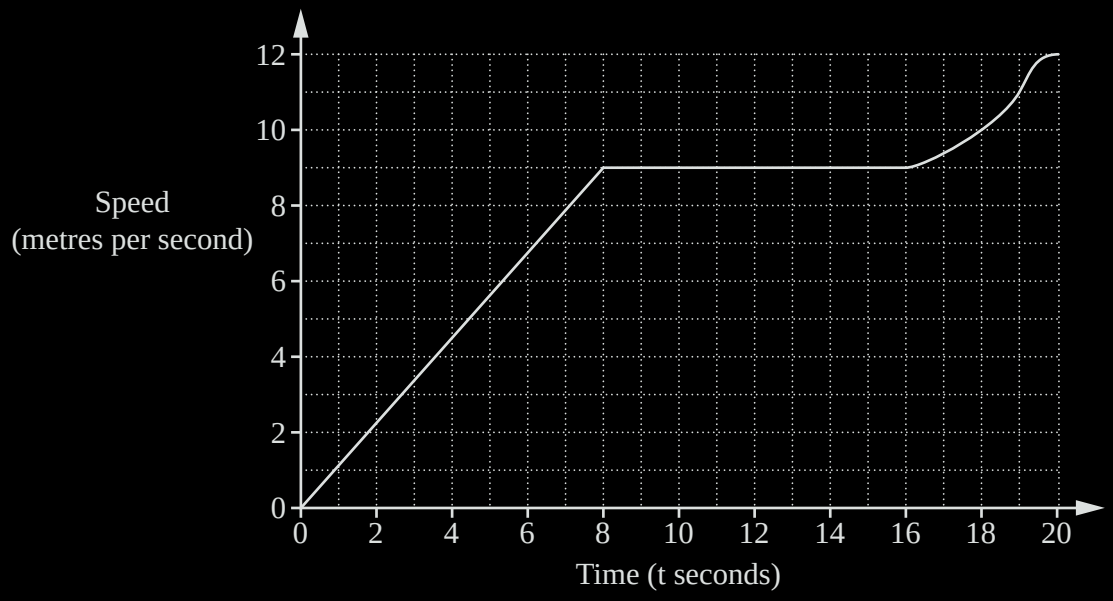
(b) Describe fully the **single** transformation that maps triangle *P* onto triangle *Q*.

Answer (b) .....

..... [2]

(c) Find the matrix representing the transformation that maps triangle *Q* onto triangle *R*.

Answer (*c*) $\begin{pmatrix} & \\ & \end{pmatrix}$ [1]



(a) Calculate the distance travelled in the first 16 seconds.

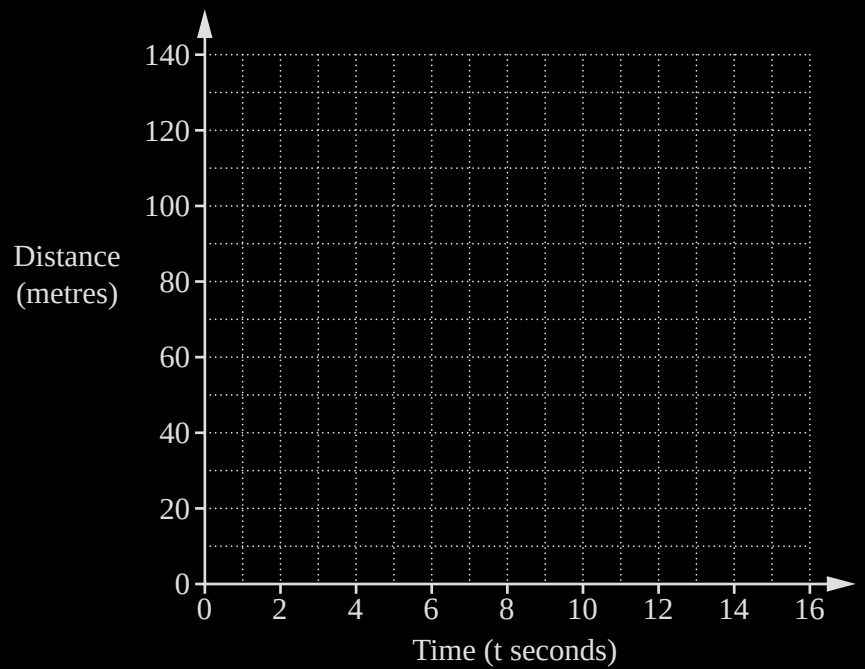
Answer (*a*) ..... m [1]

(b) By drawing a tangent, find the acceleration of the cyclist when $t = 18$.

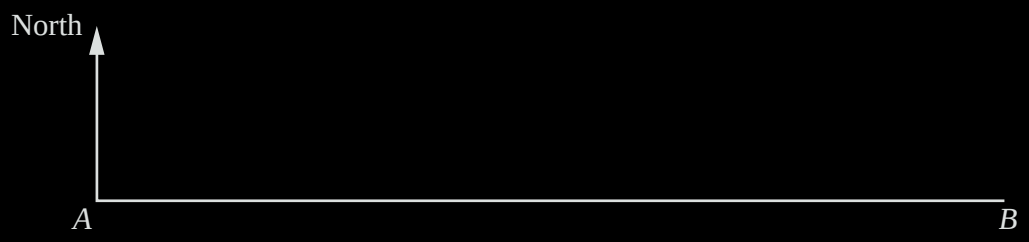
Answer (*b*) ..... m/s^2 [2]

(c) On the grid in the answer space, sketch the distance-time graph for the first 16 seconds of the cyclist's journey.

Answer (c)



[2]



(a) A third radio mast, *C*, is north of the line *AB*.
It is 40 km from *A* and 50 km from *B*.

Using ruler and compasses, construct triangle *ABC*. [2]

(b) A house *D*, inside the triangle, is more than 35 km from *B* and closer to *B* than to *A*.

Shade the region on your diagram that represents the possible positions of the house *D*. [3]